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Drive module assembly

Abstract

A drive module assembly for use in a vehicle includes a housing defining a housing interior and an electric machine. The electric machine includes a rotor and a stator. The drive module assembly includes a first input shaft, a second input shaft, a first output shaft, a second output shaft, and a differential disposed downstream of at least the rotor. The drive module assembly also includes a gearset, a clutch, and a park lock. The gearset, the clutch, and the park lock are disposed downstream of the differential such that the gearset and the clutch are configured receive rotational torque from the differential through the first and second output shafts.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and all the benefits of U.S. Provisional Application No. 63/535,553, filed Aug. 30, 2023, and U.S. Provisional Application No. 63/671,997, filed Jul. 16, 2024, which are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION
1. Field of the Disclosure
(1) The subject disclosure generally relates to a drive module assembly and, in particular, a drive module assembly for use in a vehicle.
2. Description of the Related Art
(2) Conventional drive module assemblies include an electric machine configured to deliver rotational torque to wheels of a vehicle. To help deliver rotational torque to wheels of a vehicle, conventional drive module assemblies include gears and a differential to ultimately deliver the rotational torque from the electric machine to wheels of the vehicle to allow the vehicle to maneuver.
(3) As hybrid vehicles and battery electric vehicles become more prevalent, the need for efficient and reliable drive module assemblies has become increasingly important. One of the main challenges in designing drive module assemblies for hybrid vehicles and battery electric vehicles is achieving high efficiency while maintaining a compact size and low weight. Hybrid vehicles and battery electric vehicles often have limited space available for the drive module assembly, and any added weight can reduce the vehicle's range and performance.
(4) In recent years, advances in electric motor technology and power electronics have led to the development of more compact and efficient drive module assemblies. However, in recent years, due to this development of more compact drive module assemblies, there have been challenges with integrating parking locks within the drive module assemblies along with providing torque vectoring and/or limited slip capabilities, if desired. Therefore, there still remains a need for a drive module assembly being lighter and more compact, all while having improved efficiency and increased performance while addressing the deficiencies set forth above.
SUMMARY OF THE INVENTION
(5) A drive module assembly for use in a vehicle includes a housing defining a housing interior and an electric machine. The electric machine includes a rotor disposed in the housing interior and

extending along a rotor axis, and a stator disposed about the rotor with respect to the rotor axis such that the stator surrounds the rotor. The drive module assembly includes a first input shaft rotatably coupled to the rotor and disposed downstream of the rotor such that the first input shaft is configured to receive rotational torque from the rotor, and a second input shaft rotatably coupled to the rotor and disposed downstream of the rotor such that the second input shaft is configured to receive rotational torque from the rotor. The drive module assembly also includes a first output shaft rotatably coupled to the first input shaft and the rotor and disposed downstream of the rotor and the first input shaft such that the first output shaft is configured to receive rotational torque from the rotor and the first input shaft, and a second output shaft rotatably coupled to the second input shaft and the rotor and disposed downstream of the rotor and the second input shaft such that the second output shaft is configured to receive rotational torque from the rotor and the second input shaft. The drive module assembly further includes a differential rotatably coupled to the rotor, the first and second input shafts, and the first and second output shafts. The differential is disposed downstream of at least the rotor such that the differential is configured to receive rotational torque from the rotor. The drive module assembly also includes a gearset coupled to the first and second output shafts for selectively rotatably coupling the first and second output shafts to one another. The drive module assembly additionally includes a clutch rotatably coupled to one of the first and second output shafts and to the gearset. The clutch is actuatable between a first clutch position where the first and second output shafts are uncoupled from one another, and a second clutch position where the first and second output shafts are rotatably coupled to one another through the gearset. The drive module assembly further includes a park lock configured to move between a lock position where the park lock is coupled to the gearset such that the gearset and the first and second output shafts are locked, and an unlock position where the park lock is uncoupled from the gearset such that the gearset is unlocked. The gearset, the clutch, and the park lock are disposed downstream of the differential such that the gearset and the clutch are configured to receive rotational torque from the differential through the first and second output shafts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other advantages of the present invention will be readily appreciated, as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings.

(2) FIG. **1** is a cross-sectional view of a drive module assembly including a housing, an electric machine including a rotor and a stator, a differential, a first input shaft, a second input shaft, a first counter shaft, a second counter shaft, a first output shaft, and a second output shaft.

(3) FIG. **2** is a graph illustrating a gear ratio versus a 1.sup.st drive gear base diameter.

(4) FIG. **3** is a graph illustrating a gear ratio versus a center distance between an input axis of the first and second input shafts and a counter shaft axis of the first and second counter shafts.

(5) FIG. **4** is an exemplary flow diagram of reducing the size of the drive module assembly.

(6) FIG. **5** is a cross-sectional view of another embodiment of the drive module assembly, with the drive module assembly including a gearset, a clutch, and a park lock.

(7) FIG. **6** is a cross-sectional view of the drive module assembly of FIG. **5**.

(8) FIG. **7** is a schematic of one embodiment of the gearset, the clutch, and the park lock, where the ring gear is rotatably coupled to the second output shaft, the park lock is engageable with the carrier, the clutch is disposed between the sun gear and the carrier, and the clutch is adjacent the first output shaft.

(9) FIG. **8** is a schematic of one embodiment of the gearset, the clutch, and the park lock, where the ring gear is rotatably coupled to the second output shaft, the park lock is engageable with the

carrier, the clutch is disposed between the sun gear and the carrier, and the clutch is adjacent the second output shaft.
(10) FIG. **9** is a schematic of one embodiment of gearset, the clutch, and the park lock, where the carrier is rotatably coupled to the second output shaft, the park lock is engageable with the ring gear, and the clutch is disposed between the sun gear and the carrier.
(11) FIG. **10** is a schematic of one embodiment of the gearset, the clutch, and the park lock, where the carrier is rotatably coupled to the second output shaft, the park lock is engageable with the ring gear, and the clutch is disposed between the sun gear and the ring gear.
(12) FIG. **11** is a perspective view of the gearset and the clutch.
(13) FIG. **12** is another embodiment of the drive module assembly, with the parking lock and the gearset disposed between and rotatably coupled to the first and second output shafts.
(14) FIG. **13** is another embodiment of the drive module assembly, with the clutch disposed between and rotatably coupled to the first and second output shafts.
DETAILED DESCRIPTION
(15) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a drive module assembly **10** for use in a vehicle, is shown in a cross-sectional view in FIG. **1**. The vehicle may include an internal combustion engine and, therefore, be a hybrid vehicle. The vehicle may also be a battery electric vehicle that is free of an internal combustion engine. It is to be appreciated that the vehicle may include one or more of the drive module assembly **10** described below, such as two drive module assemblies.
(16) The drive module assembly **10** includes a housing **14**, which may be one or more components, defining a housing interior **16** and an electric machine **18**. The electric machine **18** includes a rotor **20** disposed in the housing interior **16** and extending along a rotor axis RA, and a stator **22** disposed about the rotor **20** with respect to the rotor axis RA such that the stator **22** surrounds the rotor **20**. The stator **22** may be commonly referred to as a lamination stack.
(17) The drive module assembly **10** additionally includes a differential **24** rotatably coupled to the rotor **20**, a first input shaft **30** rotatably coupled to the differential **24**, and a second input shaft **32** rotatably coupled to the differential **24**. The differential **24** may be directly rotatably coupled to the rotor **20** or may have an intermediate component or components between the rotor **20** and the differential **24**. When the differential **24** is directly coupled to the rotor **20**, the differential **24** may be directly coupled with the lamination steel of the rotor **20**. Similarly, the differential **24** may be directly coupled to the first input shaft **30** and the second input shaft **32** or may have an intermediate component or components between the differential **24** and the first and second input shafts **30**, **32**. The differential **24** may include a differential pinion gear **26** and a differential side gear **28**.
(18) The rotor **20** may define a rotor interior **78** with the differential **24** disposed in the rotor interior **78**. In such embodiments, the rotor **20** may be commonly referred to as a hollow rotor shaft. The first and second input shafts **30**, **32** may be disposed in the rotor interior **78** when the rotor **20** is a hollow rotor shaft.
(19) The drive module assembly **10** further includes a first counter shaft **38** rotatably coupled to the first input shaft **30**, a second counter shaft **40** rotatably coupled to the second input shaft **32**, a first output shaft **46** rotatably coupled to the first counter shaft **38**, and a second output shaft **48** rotatably coupled to the second counter shaft **40**. The first and second output shafts **38**, **48** are typically configured to provide rotational torque to wheels of the vehicle. The first output shaft **46** may include a first output spline **70** and the second output shaft **48** may include a second output spline **72**.
(20) The differential **24** is configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32**. The first counter shaft **38** may have a first counter drive gear **42** that is rotatably coupled to the first counter shaft **38** and a second counter drive gear **44** that is rotatably coupled to

the second counter shaft **40**. The first counter drive gear **42** may be integral with the first counter shaft **38** and the second counter drive gear **44** may be integral with the second counter shaft **40**.

(21) Having the differential **24** configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** offers several advantages. First, the differential **24** is able to split the rotational torque received from the rotor **20** and is able to divide the rotational torque between the first input shaft **30** and the second input shaft **32**. In other words, the first input shaft **30** may receive one half of the rotational torque from the electric machine **18** and the second input shaft **32** may receive the other half of the rational torque from the electric machine **18**. Because the first input shaft **30** receives half of the rotational torque from the electric machine **18** and the second input shaft **32** receives the other half of the rotational torque from the electric machine **18**, the size (diameter) of the first and second input shafts **30**, **32** may be reduced, as described in further detail below. Second, having the differential **24** configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** allows for a compact design of the drive module assembly **10** all while enabling higher gear ratios, as described in further detail below.

(22) The drive module assembly **10** may include a first counter driven gear **62** rotatably coupled to the first input shaft **30** and the first counter shaft **38** and configured to deliver rotational torque from the first input shaft **30** to the first counter shaft **38**, and a second counter driven gear **64** rotatably coupled to the second input shaft **32** and the second counter shaft **40** and configured to deliver rotational torque from the second input shaft **32** to the second counter shaft **40**.

(23) The drive module assembly **10** may further include a first output gear **66** rotatably coupled to the first counter shaft **38** and the first output shaft **46** and configured to deliver rotational torque from the first counter shaft **38** to the first output shaft **46**, and a second output gear **68** rotatably coupled to the second counter shaft **40** and the second output shaft **48** and configured to deliver rotational torque from the second counter shaft **40** to the second output shaft **48**.

(24) The drive module assembly **10** may include a first bearing **50** coupled to the first input shaft **30** and configured to support rotation of the first input shaft **30**, and a second bearing **52** coupled to the second input shaft **32** and configured to support rotation of the second input shaft **32**. The first input shaft **30** has a first input diameter **54**, the second input shaft has a second input diameter **56**, the first bearing **50** has a first bearing diameter **58**, and the second bearing **52** has a second bearing diameter **60**. Having the differential **24** configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** allows the first and second input diameters **54**, **56** and the first and second bearing diameters **58**, **60** to be reduced in size. For example, the first input diameter **54** may be less than 32 millimeters, and the second input diameter **56** may be less than 32 millimeters. In another embodiment, the first input diameter **54** may be less than 29 millimeters, and the second input diameter **56** may be less than 29 millimeters. In another embodiment, the first input diameter **54** may be less than 26 millimeters, and the second input diameter **56** may be less than 26 millimeters. In another embodiment, the first input diameter **54** may be less than 23 millimeters, and the second input diameter **56** may be less than 23 millimeters. In another embodiment, the first input diameter **54** may be less than 20 millimeters, and the second input diameter **56** may be less than 20 millimeters. In another embodiment, the first input diameter **54** may be less than 32 millimeters, and the second input diameter **56** may be less than 32 millimeters. As an additional example, the first bearing diameter **58** may be less than 43 millimeters, and the second bearing diameter **60** may be less than 43 millimeters. In another embodiment, the first bearing diameter **58** may be less than 41 millimeters, and the second bearing diameter **60** may be less than 41 millimeters. In another embodiment, the first bearing diameter **58** may be less than 39 millimeters, and the second bearing diameter **60** may be less than 39 millimeters. In another embodiment, the first bearing diameter **58** may be less than 37 millimeters, and the second bearing diameter **60** may

be less than 37 millimeters. In one embodiment, the first input diameter **54** and the second input diameter **56** are 19 millimeters and the first bearing diameter **58** and the second bearing diameter **60** are 35 millimeters.

(25) Typically, the first and second bearing diameters **58**, **60** is dependent on the first and second input diameters **54**, **56**, respectively. In other words, depending on the first and second input diameters **54**, **56**, the first and second bearing diameters **58**, **60** are adjusted. Having the differential **24** configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** allows the first and second input diameters **54**, **56** to be reduced and, in turn, the first and second bearing diameters **58**, **60** to be reduced, which ultimately allows for a higher gear ratio due to the reduced space occupied by the first and second input shafts **30**, **32** and the first and second bearings **50**, **52** in the drive module assembly **10**. A reduced diameter in the first and second bearing diameters **58**, **60** reduces the rotating pitch diameter speed of the first and second bearings **50**, **52**, which allows for greater selection in bearings to use in the drive module assembly **10**.

(26) The first input shaft **30** may have a first input drive gear **34** that is rotatably coupled to the first input shaft **30** and a second input drive gear **36** that is rotatably coupled to the second input shaft **32**. The first input drive gear **34** may be integral with the first input shaft **30** and the second input drive gear **36** may be integral with the second input shaft **32**. The first input drive gear **34** may have a first drive gear diameter **74** and the second input drive gear **36** may have a second drive gear diameter **76**. As described above, having the first input diameter **54** of the first input shaft **30** being decreased in size and the second input diameter **56** of the second input shaft **32** being decreased in size allows for decreased first and second bearing diameters **58**, **60**, but also allows for decreased first and second drive gear diameters **74**, **76**. Having the decreased first and second drive gear diameters **74**, **76** allows for a higher gear ratio of the first and/or second input drive gears **34**, **36**, which is illustrated in FIG. **2**. The base diameter in FIG. **2** refers to the first and/or second drive gear diameters **74**, **76**. Similarly, FIG. **3** illustrates a gear ratio of the first and/or second input drive gears **34**, **36** versus a center distance. The center distance in the graph of FIG. **3** represents the distance between the input axis IA and the counter shaft axis CSA. To this end, the first and second input diameters **54**, **56** impact the first and second drive gear diameters **74**, **76** because the first and second drive gear diameters **74**, **76** cannot be less than the first and second input diameters **54**, **56**, and, therefore, due to reduced diameter of the first and second input diameters **54**, **56**, the first and second drive gear diameters **74**, **76** may be reduced, which allows for a higher gear ratio. A higher gear ratio allows for higher torque applications of the drive module assembly **10**. For example, the drive module assembly **10** may have a peak torque of about 250 Nm, 300 Nm, 350 Nm, 400 Nm, 450 Nm, 500 Nm, 550 Nm, 600 Nm, 650 Nm, or more, a peak speed (such as the first and second input shafts **30**, **32**) of greater than 20,000 RPM, greater than 21,000 RPM, greater than 22,000 RPM, and greater than 23,000 RPM or more, and a gear ratio of 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 or greater.

(27) The first and second input shafts **30**, **32** may extend along an input axis IA, the first and second counter shafts **38**, **40** may extend along a counter shaft axis CSA, and the first and second output shafts **46**, **48** extend along an output axis OA. In one embodiment, the input axis IA, the counter shaft axis CSA, and the output axis OA are parallel to one another. The input axis IA, the counter shaft axis CSA, and the output axis OA may be offset from one another. In one embodiment, the counter shaft axis CSA is disposed between the input axis IA and the output axis OA.

(28) In addition to the advantages set forth above, having the differential **24** configured to receive rotational torque from the electric machine **18** and configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** allows the differential **24** to be smaller. This reduces the torque input to the geartrains, such as the first and second input drive gears **34**, **36**, the first and second counter driven gears **62**, **64**, and the first and second output gears

66, **68**. Having a reduction in size of the differential **24** based on being configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32** provides the opportunity, if desired, to reduce the first input diameter **54** of the first input shaft **30** and/or the second input diameter **56** of the second input shaft **32**. Additionally, when the first input diameter **54** of the first input shaft **30** and the second input diameter **56** of the second input shaft **32** is reduced, a diameter of the first and second input drive gears **34**, **36** may also be reduced. Due to a reduction in the diameter of the first and second input drive gears **34**, **36**, a distance defined between the input axis IA and the counter shaft axis CSA may also be reduced. Furthermore, when the first input diameter **54** of the first input shaft **30** and the second input diameter **56** of the second input shaft **32** is reduced, the first and second bearing diameters **58**, **60** of the first and second bearings **50**, **52** may also be reduced. Reducing the first and second bearing diameters **58**, **60** then allows the first and second input shafts **30**, **32** to rotate faster, which, in turn, allows the electric machine **18** to rotate the rotor **20** faster. This is summarized in FIG. **4**.

(29) With reference to FIGS. **5-11**, another embodiment of the drive module assembly **10** is shown. Although not explicitly shown in FIG. **5**, the drive module assembly **10** includes the housing **14**, which may be one or more components, defining the housing interior **16**, as illustrated in FIG. **1**. With reference again to FIGS. **5** and **6**, the drive module assembly **10** includes an electric machine **18** including the rotor **20** disposed in the housing interior **16** and extending along the rotor axis RA, and the stator **22** (not explicitly shown in FIGS. **5-11**, see FIG. **1**) disposed about the rotor **20** with respect to the rotor axis RA such that the stator **22** surrounds the rotor **20**. The stator **22** may be commonly referred to as a lamination stack.

(30) The drive module assembly **10** further includes the first input shaft **30** rotatably coupled to the rotor **20** and disposed downstream of the rotor **20** such that the first input shaft **30** is configured to receive rotational torque from the rotor **20**. The drive module assembly **10** additionally includes the second input shaft **32** rotatably coupled to the rotor **20** and disposed downstream of the rotor **20** such that the second input shaft **320** is configured to receive rotational torque from the rotor **20**. In the context of this disclosure, it is to be appreciated that when a first component is downstream of a second component that the second component receives rotational torque from the first component. For example, when the first and second input shafts **30**, **32** are downstream of the rotor **20**, the first and second input shafts **30**, **32** receive rotational torque from the rotor **20**, either directly (i.e., components are directly rotatably coupled to one another) or indirectly (i.e., there may be an intermediate component between the first component and the second component). It is also to be appreciated in the context of this disclosure that the first input shaft **30** and second input shaft **32** may be referred to as a first shaft and a second shaft, respectively. The first input shaft **30** and the second input shaft **32** provide rotational torque to the first and second output shafts **46**, **48**, respectively. The first input shaft **30** and second input shaft **32** receive rotational torque from the electric machine **18** and, therefore, may be referred to as output shafts with respect to the electric machine **18**.

(31) The drive module assembly **10** also includes the first output shaft **46** rotatably coupled to the first input shaft **30** and the rotor **20** and disposed downstream of the rotor **20** and the first input shaft **30** such that the first output shaft **46** is configured to receive rotational torque from the rotor **20** and the first input shaft **30**. The drive module assembly **10** additionally includes the second output shaft **48** rotatably coupled to the second input shaft **32** and the rotor **20** and disposed downstream of the rotor **20** and the second input shaft **32** such that the second output shaft **48** is configured to receive rotational torque from the rotor **20** and the second input shaft **32**.

(32) The drive module assembly **10** further includes the differential **24** rotatably coupled to the rotor **20**, the first and second input shafts **30**, **32**, and the first and second output shafts **46**, **48**. The differential **24** is disposed downstream of at least the rotor **20** such that the differential **24** is configured to receive rotational torque from the rotor **20**. In other words, the first and second input shafts **30**, **32** and the first and second output shafts **46**, **48** may be downstream of the differential

24, as shown in FIGS. **5** and **6**, or the differential **24** may be downstream of the first and second input shafts **30**, **32** and the first and second output shafts **46**, **48**. In one embodiment, the differential **24** is configured to receive rotational torque from the electric machine **18** and is configured to transmit rotational torque from the electric machine **18** to the first and second input shafts **30**, **32**. The differential **24** may be directly coupled to the rotor **20**. The differential **24** may be directly coupled to the first and second input shafts **30**, **32**.

(33) The drive module assembly **10** in the embodiment of FIGS. **5-11** includes a gearset **80** coupled to the first and second output shafts **46**, **48** for selectively rotatably coupling the first and second output shafts **46**, **48** to one another. As described in further detail below, the gearset **80** may be any suitable gearset, such as a planetary gearset. The drive module assembly **10** further includes a clutch **82** rotatably coupled to one of the first and second output shafts **46**, **48** and to the gearset **80**. The clutch **82** is actuatable between a first clutch position where the first and second output shafts **46**, **48** are uncoupled from one another, which may be alternatively referred to as an open clutch position, and a second clutch position where the first and second output shafts **46**, **48** are rotatably coupled to one another through the gearset, which may be alternatively referred to as a closed clutch position. To this end, the clutch **82** may be used for torque vectoring between the first and second output shafts **46**, **48**. Alternatively, or additionally, the clutch **82** may be used as a limited slip differential. The drive module assembly **10** includes a park lock **84**, such as a park pawl, configured to move between a lock position where the park lock **84** is coupled to the gearset **80** such that the gearset **80** and the first and second output shafts **46**, **48** are locked, and an unlock position where the park lock **84** is uncoupled from the gearset **80** such that the gearset **80** is unlocked.

(34) The gearset **80**, the clutch **82**, and the park lock **84** are disposed downstream of the differential **24** such that the gearset **80** and the clutch **82** are configured to receive rotational torque from the differential **24** through the first and second output shafts **46**, **48**. Having the gearset **80**, the clutch **82**, and the park lock **84** disposed downstream of the differential **24** such that the gearset **80** and the clutch **82** are configured to receive rotational torque from the differential **24** through the first and second output shafts **46**, **48** offers several advantages. First, having the gearset **80**, the clutch **82**, and the park lock **84** disposed downstream of the differential **24** such that the gearset **80** and the clutch **82** are configured to receive rotational torque from the differential **24** through the first and second output shafts **46**, **48** allows for a compact design. Second, having the gearset **80**, the clutch **82**, and the park lock **84** disposed downstream of the differential **24** such that the gearset **80** and the clutch **82** are configured to receive rotational torque from the differential **24** through the first and second output shafts **46**, **48** allows both the clutch **82** and the park lock to have a single actuation system, such as hydraulic actuation.

(35) The drive module assembly **10** may include the first counter shaft **38** rotatably coupled to the first input shaft **30**, and the second counter shaft **40** rotatably coupled to the second input shaft **32**, with the first output shaft **46** being rotatably coupled to the first counter shaft **38** and the second output shaft **48** being rotatably coupled to the second counter shaft **40**.

(36) The first output shaft **46** may extend along a first output axis OA**1** and the second output shaft **48** may extend along a second output axis OA**2**. The first output axis OA**1** and the second output axis OA**2** may be parallel with one another. In some embodiments, the first output axis OA**1** and second output axis OA**2** are coaxial with one another.

(37) Typically, the gearset **80** and the clutch **82** are disposed about at least one of the first output axis OA**1** and the second output axis OA**2**. However, the gearset **80** and the clutch **82** may be disposed about both the first output axis OA**1** and the second output axis OA**2**.

(38) The gearset **80** may be further defined as a planetary gearset including a carrier **86**, a sun gear **88**, a plurality of planet gears **90**, and a ring gear **92**. As shown in FIGS. **5** and **6**, the sun gear **88** is typically rotatably coupled to the first output shaft **46** and, in some embodiments, directly coupled to the first output shaft **46**.

(39) As illustrated in FIGS. **7** and **8**, in one embodiment, the ring gear **92** is rotatably coupled to the second output shaft **48** (e.g. through an intermediate shaft **96** shown in FIG. **6** or directly coupled as shown in FIG. **5**), the park lock **84** is engageable with the carrier **86**, and the clutch **82** is disposed between and rotatably coupled to the sun gear **88** and the carrier **86**. As shown in FIG. **7**, the clutch **82** is adjacent the first output shaft **46**, and in FIG. **8** the clutch **82** is adjacent the second output shaft **48**. With reference to FIG. **9**, the carrier **86** is rotatably coupled to the second output shaft **48**, the park lock **84** is engageable with the ring gear **92**, and the clutch **82** is disposed between and rotatably coupled to the sun gear **88** and the carrier **86**. With reference to FIG. **10**, the carrier **86** is rotatably coupled to the second output shaft **48**, the park lock **84** is engageable with the ring gear **92**, and the clutch **82** is disposed between and rotatably coupled to the sun gear **88** and the ring gear **92**.
(40) With reference to FIG. **11**, the carrier **86** may include at least one groove **94** configured to receive the park lock **84** when the park lock **84** is in the lock position. The at least one groove **94** may be any suitable configuration, such as a gear shown in FIG. **11**.
(41) It is to be appreciated, such as shown in FIG. **12**, that the parking lock **84** and the gearset **80** disposed between and rotatably coupled to the first and second output shafts **46**, **48** without the clutch **82**. It is also to be appreciated, as shown in FIG. **13**, that the clutch **82** may be disposed between and rotatably coupled to the first and second output shafts **46**, **48** without the park lock **84**.
(42) The rotor **20** may define the rotor interior **78** with the differential **24** disposed in the rotor interior **78**. In such embodiments, the rotor **20** may be commonly referred to as a hollow rotor shaft. The first and second input shafts **30**, **32** may be disposed in the rotor interior **78** when the rotor **20** is a hollow rotor shaft.
(43) The vehicle may include an internal combustion engine and, therefore, be a hybrid vehicle. The vehicle may also be a battery electric vehicle that is free of an internal combustion engine. It is to be appreciated that the vehicle may include one or more of the drive module assembly **10** described above, such as two drive module assemblies.

Claims

1. A drive module assembly for use in a vehicle, said drive module assembly comprising: a housing defining a housing interior; an electric machine, comprising, a rotor disposed in said housing interior and extending along a rotor axis, and a stator disposed about said rotor with respect to said rotor axis such that said stator surrounds said rotor; a first input shaft rotatably coupled to said rotor and disposed downstream of said rotor such that said first input shaft is configured to receive rotational torque from said rotor; a second input shaft rotatably coupled to said rotor and disposed downstream of said rotor such that said second input shaft is configured to receive rotational torque from said rotor; a first output shaft rotatably coupled to said first input shaft and said rotor and disposed downstream of said rotor and said first input shaft such that said first output shaft is configured to receive rotational torque from said rotor and said first input shaft; a second output shaft rotatably coupled to said second input shaft and said rotor and disposed downstream of said rotor and said second input shaft such that said second output shaft is configured to receive rotational torque from said rotor and said second input shaft; a differential rotatably coupled to said rotor, said first and second input shafts, and said first and second output shafts, wherein said differential is disposed downstream of at least said rotor such that said differential is configured to receive rotational torque from said rotor; a gearset coupled to said first and second output shafts for selectively rotatably coupling said first and second output shafts to one another; a clutch rotatably coupled to one of said first and second output shafts and to said gearset, wherein said clutch is actuatable between a first clutch position where said first and second output shafts are uncoupled from one another, and a second clutch position where said first and second output shafts are rotatably coupled to one another through said gearset; and a park lock configured to move between

a lock position where said park lock is coupled to said gearset such that said gearset and said first and second output shafts are locked, and an unlock position where said park lock is uncoupled from said gearset such that said gearset is unlocked; wherein said gearset, said clutch, and said park lock are disposed downstream of said differential such that said gearset and said clutch are configured to receive rotational torque from said differential through said first and second output shafts.
2. The drive module assembly as set forth in claim 1, wherein said first output shaft extends along a first output axis and said second output shaft extends along a second output axis, and wherein said first output axis and said second output axis are parallel with one another.
3. The drive module assembly as set forth in claim 2, wherein said first output axis and said second output axis are coaxial with one another.
4. The drive module assembly as set forth in claim 2, wherein said gearset and said clutch are disposed about at least one of said first output axis and said second output axis.
5. The drive module assembly as set forth in claim 4, wherein said gearset and said clutch are disposed about both of said first output axis and said second output axis.
6. The drive module assembly as set forth in claim 1, wherein said gearset is further defined as a planetary gearset comprising a carrier, a sun gear, a plurality of planet gears, and a ring gear.
7. The drive module assembly as set forth in claim 6, wherein said sun gear is rotatably coupled to said first output shaft.
8. The drive module assembly as set forth in claim 7, wherein said ring gear is rotatably coupled to said second output shaft, wherein said park lock is engageable with said carrier, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said carrier.
9. The drive module assembly as set forth in claim 7, wherein said carrier is rotatably coupled to said second output shaft, wherein said park lock is engageable with said ring gear, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said carrier.
10. The drive module assembly as set forth in claim 7, wherein said carrier is rotatably coupled to said second output shaft, wherein said park lock is engageable with said ring gear, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said ring gear.
11. The drive module assembly as set forth in claim 6, wherein said ring gear is rotatably coupled to said second output shaft, wherein said park lock is engageable with said carrier, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said carrier.
12. The drive module assembly as set forth in claim 11, wherein clutch is adjacent said first output shaft.
13. The drive module as set forth claim 11, wherein said clutch is adjacent said second output shaft.
14. The drive module assembly as set forth in claim 6, wherein said carrier is rotatably coupled to said second output shaft, wherein said park lock is engageable with said ring gear, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said carrier.
15. The drive module assembly as set forth in claim 6, wherein said carrier is rotatably coupled to said second output shaft, wherein said park lock is engageable with said ring gear, and wherein said clutch is disposed between and rotatably coupled to said sun gear and said ring gear.
16. A vehicle comprising said drive module assembly as set forth in claim 1.
17. A vehicle comprising said drive module as set forth in claim 1, wherein said vehicle comprises an internal combustion engine.
18. A vehicle comprising said drive module as set forth in claim 1, wherein said vehicle is free of an internal combustion engine.